

Bachelor Project



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F3

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Design of a platform for 2D LiDaR

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Field of study: Electronics and Communications

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Acknowledgements

Děkuji ČVUT, že mi je tak dobrou *alma mater*.

Declaration

Prohlašuji, že jsem předloženou práci vypracoval samostatně, a že jsem uvedl veškerou použitou literaturu.

V Praze, 13. November 2025

Abstract

There are currently many advanced LiDAR sensors available for 3D spatial scanning [*insert citation]. However, most of these sensors are too expensive for the average user. The most affordable commercial option is the Unitree L1, which costs \$ 250 [*insert citation]. The aim of this miniproject is to design a platform for a 360° LiDAR sensor, which will be utilized in a subsequent bachelor's thesis to visualize a 3D image of the scanned environment.

Keywords: LiDAR, 3D reconstruction, point cloud, signal processing, mapping

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Abstrakt

V současnosti existuje spousta pokročilých lidar-senzorů určených pro 3D snímání prostoru [*insert citation*]. Většina z nich je však cenově nedostupná pro běžného uživatele. Nejvíce dostupným komerčním řešením je Unitree L1, který stojí \$ 250 [*insert citation*]. Cílem toho miniprojektu je navrhnout pro 360° lidar-senzor platformu a použít ji v navazující bakalářské práci pro vizualizaci 3D-obrazu naskenovaného prostoru.

Klíčová slova: LiDAR, 3D rekonstrukce, point cloud, zpracování signálů, mapování

Překlad názvu: Návrh platformy pro 2D LiDaR

Contents

Acronyms	3
1 Introduction	5
2 Design of a Platform	7
2.1 Existing DIY Solutions for Environmental Mapping	7
2.1.1 Using a Single-Point Light Detection and Ranging (LiDAR) module	7
2.1.2 Using a Single-Point LiDAR and a Mirror	8
2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform	9
2.2 Proposed Design of a 360° LiDAR Platform	11
2.2.1 Controlling the Platform	12
2.2.2 Choosing the Tilt Angle	13
2.2.3 Choosing a Gear Ratio for the Rotating Platform	14
3 Conclusion	15
A Bibliography	17

Figures

2.1	Implementation in project [5] using a single-point LiDAR	8
2.2	Implementation in project [6] using a single-point LiDAR and a mirror	9
2.3	Implementation in project [7] using a tilting platform	10
2.4	Simulation of an obtained point cloud using a tilting platform . .	10
2.5	Proposed platform design. θ is the tilt angle	11
2.6	Simulation of an obtained point cloud using a rotating tilted platform with two different tilt angles θ	12
2.7	NEMA 17 17HS8401 photo ¹ . . .	13

Tables

2.1	Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor ¹	13
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Acronyms

LiDAR Light Detection and Ranging

DSP Digital Signal Processing

SLAM Simultaneous Localization and Mapping



Chapter 1

Introduction

Nowadays, LiDAR sensors are widely used in various applications, including autonomous vehicles, robotics, and environmental mapping [1; 2, Ch. 4-6]. While the former two applications often require high-end and expensive LiDAR systems, environmental mapping can be effectively performed using more affordable sensors. However, even the cheapest commercial 3D LiDAR sensor, which was found during the research for this miniproject, costs \$249 [3]. The primary motivation behind this mini-project (and the subsequent bachelor's thesis) is to develop a cost-effective platform for a 360° LiDAR sensor that can be utilised for 3D spatial scanning and visualisation of the surrounding environment. A subsequent bachelor's thesis will then focus on the Digital Signal Processing (DSP) techniques for the effective processing of the acquired point cloud data and proper utilisation of open-source visualisation tools (such as Python's Open3D library [4]) to recreate a 3D scene.



Chapter 2

Design of a Platform

This chapter will firstly go over the existing DIY solutions for 360° LiDAR platforms which the author has stumbled upon during the research[too informal?], analyzing their pros and cons. After that, the proposed design will be presented, along with the chosen design decisions and their justifications.



2.1 Existing DIY Solutions for Environmental Mapping



2.1.1 Using a Single-Point LiDAR module

One of the most straightforward approaches to achieve 360° environmental mapping is to utilize a single-point LiDAR module mounted on a platform which is able to rotate and tilt. The implementation could be found in this project: [5].

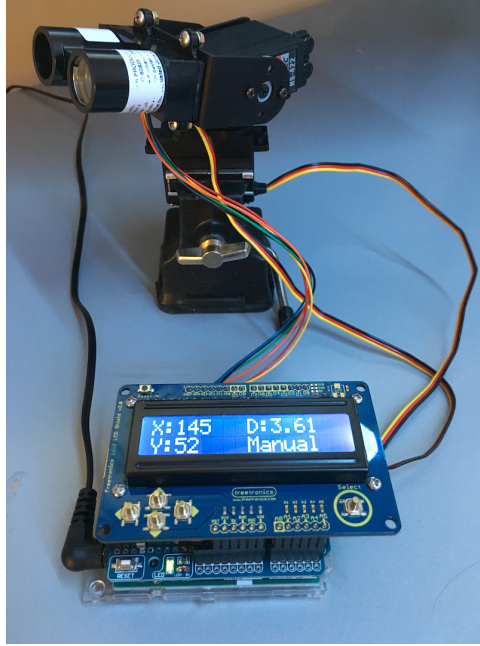


Figure 2.1: Implementation in project [5] using a single-point LiDAR

Pros

- + Single-point LiDARs are the most cost-effective among all existing types
- + Simple design

Cons

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Movement of lidar may introduce inaccuracies in distance measurements, which need to be compensated for in code if we want to assume that all points are measured from a single position in space

■ 2.1.2 Using a Single-Point LiDAR and a Mirror

The following approach was used in this project: [6]. A single-point LiDAR is placed below a rotating mirror, which reflects the laser beam horizontally. As the mirror rotates, the laser beam scans the environment in 360°. Moreover, by tilting the mirror at an angle, vertical scanning can also be achieved, allowing for the acquisition of the whole scene.

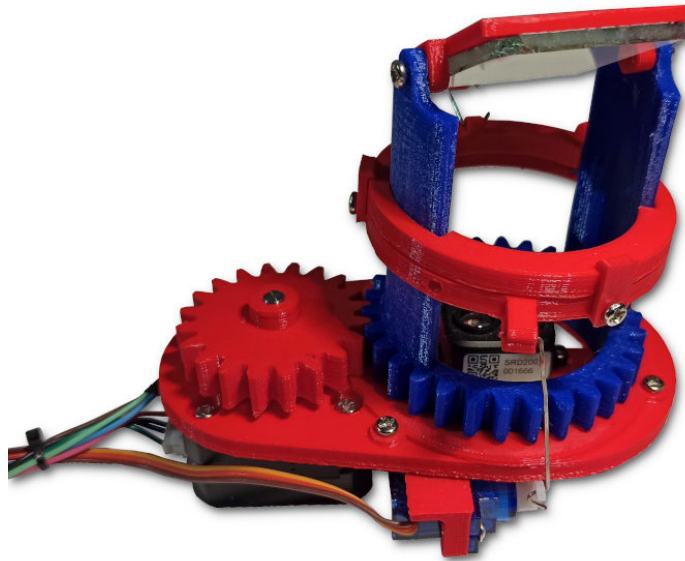


Figure 2.2: Implementation in project [6] using a single-point LiDAR and a mirror

Pros

- + Cost-effectiveness as in previous implementation
- + Simple design
- + Much easier to correct the distance measurement, since the LiDAR is stationary and the distance from it to the mirror is known.

Cons

- Slow data acquisition relative to methods leveraging 2D LiDAR's speed
- Mirror's quality may affect the accuracy of distance measurements
- Mechanical structure holding the mirror may introduce vibrations, thus affecting measurement accuracy

2.1.3 Using a 2D LiDAR Mounted on a Tilting Platform

This approach utilizes a 2D LiDAR sensor mounted on a platform that can tilt the sensor side-to-side, as demonstrated in this project: [7]. The 2D LiDAR scans the environment in a horizontal plane, while the tilting mechanism allows for vertical scanning.

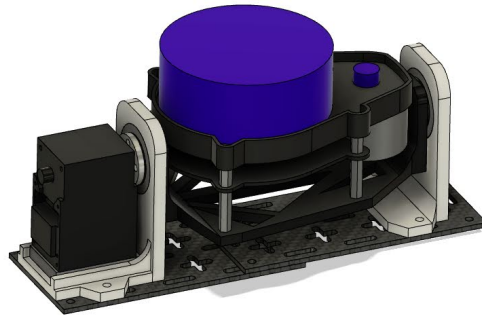


Figure 2.3: Implementation in project [7] using a tilting platform

Pros

- + Faster data acquisition compared to solutions with single-point LiDAR measurements
- + Higher resolution given by the use of a 2D LiDAR sensor
- + This design could be potentially used for Simultaneous Localization and Mapping (SLAM) applications, given the higher speed

Cons

- More expensive than single-point LiDAR solutions
- Point distribution is non-uniform due to the tilting mechanism which produces denser point clouds on the axis of tilting (see fig. 2.4)
- Mechanical design is more prone to vibrations produced by the LiDAR itself affecting measurement accuracy

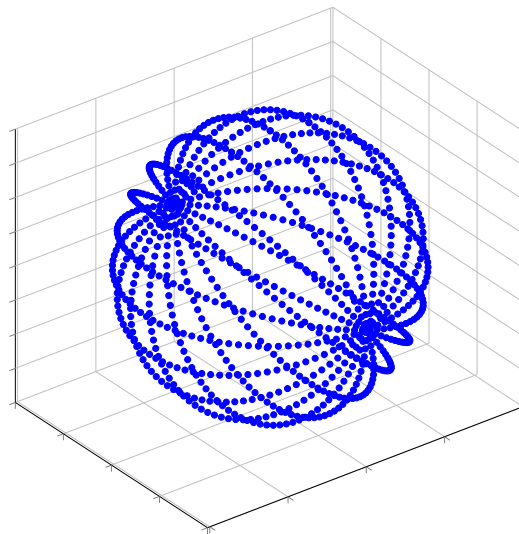


Figure 2.4: Simulation of an obtained point cloud using a tilting platform

2.2 Proposed Design of a 360° LiDAR Platform

Based on the analysis of existing DIY solutions for 360° LiDAR platforms, the proposed design for this miniproject involves using a 2D LiDAR sensor mounted on a tilted platform that is able to rotate around its axis.

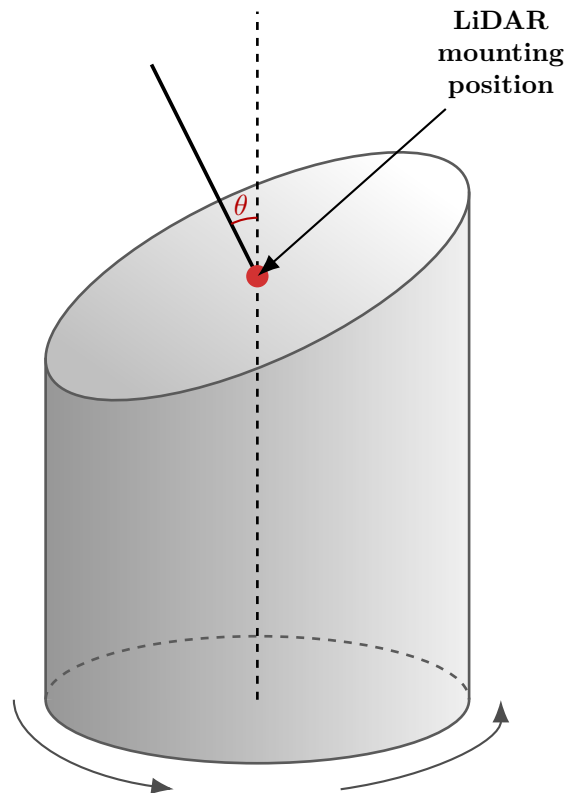


Figure 2.5: Proposed platform design. θ is the tilt angle

This design leverages the advantages of 2D LiDAR sensors, such as fast data acquisition and high resolution, while also eliminating some of the drawbacks of previous designs, namely:

- + The platform will be quite rigid, minimizing vibrations
- + Mechanical design is very simple, making it easy to build and troubleshoot
- + The resulting point cloud will be more uniformly distributed compared to the one obtained using tilting platform design presented in section 2.1.3 unless the tilt angle is extremely steep (see fig. 2.6)

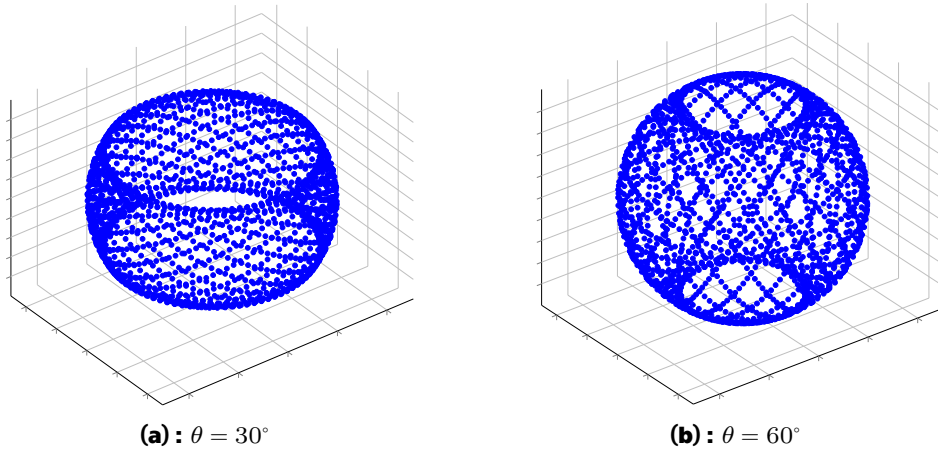


Figure 2.6: Simulation of an obtained point cloud using a rotating tilted platform with two different tilt angles θ

Such design does have its drawbacks, however. The first one is immediately apparent from fig. 2.6: there are blind spots in the point cloud where no points are measured. While these blind spots can be minimized by choosing as large a tilt angle as possible, it would reintroduce the problem of non-uniform point cloud density.

Besides that, compared to the tilting platform design presented in section 2.1.3, the proposed design would need to rotate twice as much to obtain all the full coverage of the environment, since the tilting platform scanned only within $[-90^\circ, +90^\circ]$, whereas the current design must complete a full 360° . This would lead to longer scanning times.

2.2.1 Controlling the Platform

Since we want our platform to be precisely controlled, we need to choose an appropriate motor. Stepper motors are commonly used in applications requiring precise positioning, as they can be controlled to move in small, discrete steps. For this design, a NEMA 17 17HS8401 stepper motor is used to rotate the LiDAR platform.



Figure 2.7: NEMA 17 17HS8401 photo¹

Table 2.1: Crucial parameters of the chosen NEMA 17 17HS8401 stepper motor¹

Motor Weight [Ω]	360
Holding Torque [N m]	0.5
Step Angle [°]	1.8
# of Steps per Revolution [-]	200
Rated Current [A]	1.68

For heavy load applications, it is important to consider the potential error sources of stepper motors, such as the stepper motor losing steps due to a heavy load or by driving the motor faster than its maximum speed [8, p. 13]. However, in our case, the load will be relatively light for this motor and the speed will be kept low, thus a feedback mechanism for correcting lost steps is not required.

2.2.2 Choosing the Tilt Angle

The tilt angle of the LiDAR platform is a crucial parameter that affects the size of the aforementioned blind spots in the point cloud. A larger tilt angle will result in smaller blind spots as could be seen from the simulations below.

¹Photo and parameters were taken from <https://www.laskakit.cz/en/krokovy-motor-nema-17-17hs8401-0-5nm/>

The trade-off for a larger tilt angle, however, is a more spread-out point cloud in the horizontal direction if we rotate our platform in discrete steps [see point cloud simulations below]. Without any further solutions, we would have to choose a tilt angle that balances the size of blind spots and point cloud density.

Thankfully, there is a way to achieve a more complete coverage of the environment without sacrificing point cloud density.

■ 2.2.3 Choosing a Gear Ratio for the Rotating Platform

In order to remove the need for a compromise between point cloud coverage and density while using a stepper motor to rotate the platform, we can introduce a gear mechanism into our design. Using gears with a specific gear ratios allows us to choose a tilt angle that minimizes blind spots while still maintaining a dense point cloud.



Chapter 3

Conclusion

Appendix A

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